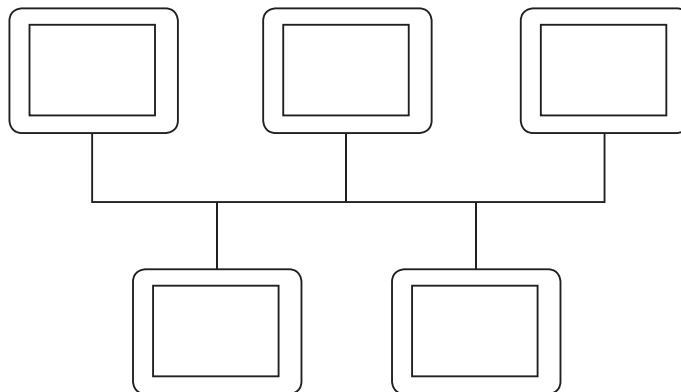


Answers to Practice Papers — 9

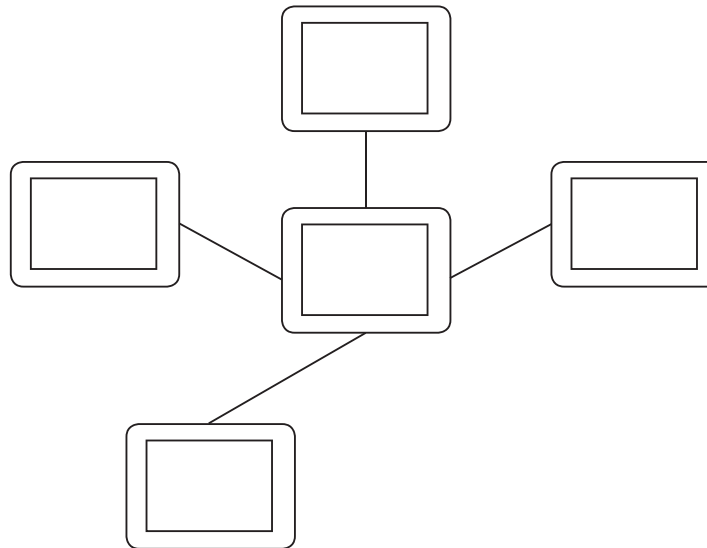
1. (c) Data Warehouse
2. (a) LIKE _____ (that's six underscore characters)
3. (b) TRUNCATE()
4. (b) Like
5. (a) Slower data transmission
6. (c) Upper
7. (d) Group by
8. (c) cookies
9. (a) True
10. (a) URL
11. (d) 0 1 0 1
12. (a) for i in list:
13. (a) "break" exits the loop entirely, while "continue" skips the current iteration and continues to the next one
14. (c) [1, 3]
15. (a) (1, 2, 3, 1, 2, 3, 1, 2, 3)
16. (a) 5
17. (a) Both A and R true and R is the correct explanation for A.
18. (c) A is True but R is False.
- 19.

```
def Sum(Count):  
    S=0  
    for I in range(1,Count+1):  
        S+=I  
    return S  
print(Sum(2)) #Function Call  
print(Sum(5))
```

20. Bus Topology:



Star Topology:



OR

(i) Wired → optical fiber

(ii) Wireless → Infrared/microwave

Wired Medium	Wireless Network
Ethernet/Twisted pair cable	Microwave
Coaxial cable	Radio waves
Optical Fibre cable	Infrared

21. Differences:

Mutable data objects	Immutable data objects
Mutable data objects can be modified after their creation.	Immutable data objects cannot be modified once they are created.
Mutable data objects are passed to functions with reference.	Immutable data objects are passed to functions with value.
Examples of mutable data objects are: lists and dictionaries.	Examples of immutable data objects are: integers, floats, strings, tuples.

22. **Distinct:** This clause is used to fetch(retrieve) unique values(non-duplicate) from the column of the table. The SELECT DISTINCT statement is used to return only distinct (different) values.

Syntax

Select distinct <column name> from <tablename>;

For example

Select distinct City from Student;

23. (i) Hypertext Markup Language
(ii) Uniform Resource Locator (URL)

24. (i) R & R & R &

max value = 1

min value = 3

OR

R

Ra

Rac

Rach

Rachn

Rachna

Rachna

Rachna S

Rachna Sa

Rachna Sag

Rachna Saga

Rachna Sagar

25. NULL. Null is undefined or unavailable or Missing value/ No value but it's not 0. If the value of the record is not known, then we can write NULL. Arithmetic or relational operators cannot work with NULL.

(a) Is Null: returns true, when column contain null value.

Example

Select * from Student where Fees is null;

(b) Is not Null: returns true, when column does not contain null value.

Example

Select * from student where Fees is not null;

OR

Primary Key

The primary key is used as a unique identifier for each record in the table. Each table supports only one primary key.

Unique Key

The unique key is also a unique identifier for records when the primary key is not present in the table. A table can have more than one unique key. For example, consider a table of employees with columns id, name, and email. To ensure that each employee has a unique email address.

26. (a) The COMMIT statement saves all the modifications made in the current. This command is used to save the changes in the database.

For example: db.commit()

(b)

```
mysql> select department, max(salary) from employees;
+-----+-----+
| department | max(salary) |
+-----+-----+
| IT         | 55000       |
+-----+-----+
1 row in set (0.03 sec)

mysql> select distinct department from employees;
+-----+
| department |
+-----+
| IT         |
| HR         |
+-----+
2 rows in set (0.04 sec)

mysql> select name from employees where name like '%sharma';
+-----+
| name          |
+-----+
| Amit Sharma   |
| Naveen Sharma |
+-----+
2 rows in set (0.00 sec)

mysql> select employee_id,name from employees where department='IT' order by salary;
+-----+-----+
| employee_id | name          |
+-----+-----+
| 1           | Amit Sharma   |
| 5           | Sonal Patel   |
| 3           | Vishal Gupta  |
+-----+-----+
3 rows in set (0.00 sec)
```

```
27. def count_W_H():
    f = open ("Country.txt", "r")
    W,H = 0,0
    r = f.read()
    for x in r:
        if x[0] == "W" or x[0] == "w":
            W=W+1
        elif x[0] == "H" or x[0] == "h":
            H=H+1
    f.close()
    print ("W or w :", W)
    print ("H or h :", H)
```

OR

```
def countwords():
    s = open("Quotes.txt","r")
    f = s.read()
    z = f.split ()
    count = 0
    for I in z:
        count = count + 1
    print ("Total number of words:", count)
```

28. (i)

count(*)		gender
3	M	
2	F	

(ii)

name	subject
Rohit	math
Sonia	science
Vikas	history
Anjali	english
Amit	geography

(iii)

name	subject
Rohit	math

29.

```
def M_SEARCH (STATES):
    for i in STATES:
        if i[0] == "m":
            print (i)
```

30.

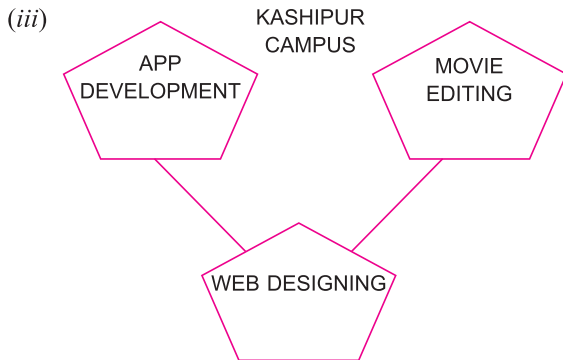
```
EMP={"EOP1":16000, "EOP2":28000, "EOP3":19000, "EOP4":15000, "EOP5":30000}
S=[]
def PUSH(EMP):
    for k in EMP:
        if EMP[k]<25000:
            S.append(k)
def POP(S):
    while S:
        print(S.pop(),end=" ")
    print("Stack Empty")
#Calling Statement
PUSH(EMP)
POP(S)
```

OR

```
Bitem={"Bread":40,"Cake":250,"Muffins":80,"Biscuits":25}
S=[]
def Push(Bitem):
    for i,j in Bitem.items():
        if j<50:
            S.append(i)
```

```
def POP(S):
    while S:
        print(S.pop())
        print("Stack Empty")
#Calling Statement
Push(Bitem)
POP(S)
```

31. (i) The server should be installed in the block with the highest number of computers. Movie Editing Block is the most appropriate to house the server as it has the maximum number of computers.
- (ii) Firewall is the device/software to be installed in Kashipur Campus to take care of data security. It acts as a barrier between the internal trusted network and the external untrusted network.



Ethernet Cable is best wired medium to economically connect the various blocks in the Kashipur.

- (iv) (a) A Switch is used to connect computer within each building. Hence, a switch must be placed in all blocks.
- (b) Repeater is not required between the blocks as the distances are less than 100 m. Since the distance between all buildings are less than 100m, repeater is not allowed.
- (v) Protocol: VoIP (Voice of Internet Protocol)

32. (a) sELCcME#Cc

(b) #Statement 1 - con1.cursor()
 #Statement 2 - mycursor.execute('select * from student where marks>75')
 #Statement 3 - mycursor.fetchall()

OR

(a) BBe@@@@HHuMMan

(b) **Statement 1:** mydb.cursor()
Statement 2: mycursor.execute("alter table student add column MARKS int")
Statement 3: mydb.commit()

33. import pickle

```
def function():
    try:
        f=open("ITEMS.dot","rb")
        while True:
            s=pickle.load(F)
            if s["cost"]<2500:
```

```

        print (s)
    except:
        F.close()
function()

```

OR

```

import pickle
def BUMPER():
    try:
        f=open("GIFTS.dot","rb")
        while true:
            c=pickle load(F)
            if c[2]=="ON DISCOUNT"
                print (c)
    except:
        F.close()
BUMPER()

```

34. (i) Select Pcode, PName, DName, City, Mobile from Distributer where Disc > 7;
(ii) Select * from Distributor where Company like "%a%";
(iii) Select Sum (Price), City from Distributor order by City;

OR

- (iii) Select PCode, PName, Company from Distributor order by Price desc;
35. (a) count word()
(b) 'para.txt', 'r'
(c) c=fread()
(d) x.split()
(e) f.close()